

END OF TERM TWO SENIOR FIVE MATHEMATICS

Paper 1

Time: 2:30mins

INSTRUCTIONS TO CANDIDATES:

- Answer **all ITEMS** in this paper
- Non-programmable scientific electronic calculators may be used.

Item 1

The National Water and Sewerage Corporation (NWSC) is designing a water distribution system for three neighboring communities in Kampala. Each community has different water requirements and infrastructure constraints. A consultant engineer working on this project wants to determine the optimal flow rates for each community. The water distribution system is modeled by the following equations

$X + 2Y + Z = 2400$ (Total available water supply in litres per minute)

$3X + Y + 3Z = 3900$ (pressure balancing equation)

$3X + 4Y + 2Z = 5100$ (Flow optimization equation)

Where X, Y and Z represent the flow rate in litres per minute in communities A, B and C respectively

The polynomial equation $P(X) = X^3 - 7X^2 + 14X - 8$ models the operational efficiency of the pumping systems

Task:

- a) Help the engineer to determine the optimal flow for each community
- b) If the community A's water requirements increase by 200 litres per minute. What adjustments should be made to the other communities' supply to maintain the system balance?
- c) Determine all the possible value of X where the efficiency of the pumping system is zero

Item 2

The Entebbe Municipal Council is planning to develop a new neighborhood with roads, residential areas, and a community center. The planning department uses a coordinate system where each unit represents 100 meters. An urban planner, needs to design the road network and determine optimal locations for the key facilities. A road P is planned to connect points A(2, 3) and B(8, 15) and another road Q will connect points C(4, 5) and D(12, 9)

Task:

- a) Help the urban planner to determine the length of the road P in kilometers
- b) Establish whether the two roads P and Q intersect, and if so, at what point and angle between the two roads will they meet.
- c) If they what to construct a curved road given by the function $x^2y + 3y^2 - 4x - 12 = 0$. Find the equation which can give a straight road passing through the curved road normally at a point (1, 2)

- d) If the parametric equations $x = \frac{1+t}{1-t}$ and $y = \frac{2t^3}{1-t}$ can give the curve in (c), use them to find its second derivative.
- e) If the area enclosed by the roads is given by $A = \int_1^2 \left(\frac{x^3 + 3x^2 - 4x}{x} \right) dx$. Find the area enclosed by the roads, A

Item 3

A renewable energy company is installing solar panels on buildings in Tororo district to maximize energy collection. They need to determine optimal installation angles based on the building's orientation and the sun's position throughout the year. The company engineer needs to know the ideal mounting angles for a solar project at a school. He finds out that the amount of solar energy collected by the panels is modeled by the function: $E(\theta) = 200\cos\theta + 300\sin\theta - 500$ where E is the energy collected in watt-hours per square meter and θ is the angle of the panels from the horizontal in radians and the shadow length L of a vertical pole of height h at a time t in hours after noon is given by: $L = h \tan\left(\frac{\pi}{12t} + \frac{\pi}{4}\right)$

Tasks:

- If the school building has a roof with a pitch of 35° facing south. If the optimal angle for solar panel is 15° relative to the roof surface, determine the absolute angle of the panels from the horizontal
- Establish the value of θ that maximizes energy collection.
- At what time will the shadow be equal to the height of the pole.
- If the engineer had an equation given by $\sin(\theta + 20^\circ) = p \sin(\theta - 20^\circ)$. Show the engineer that the same equation can be written in the form $\tan\theta = \left(\frac{p+1}{p-1}\right) \tan 20^\circ$

END

“Things are harder tomorrow, better done today”